

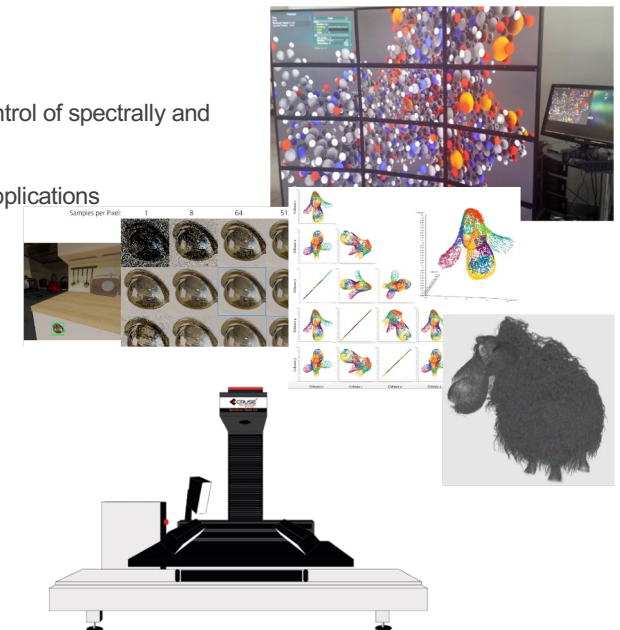
Facultiy DCSM
at RheinMain University of
Applied Sciences
HCPM 2026, NAIST

Prof. Dr.-Ing. Martin Weier
08.09.2028

Introduction

Martin Weier

- Background:
 - Studies: Computer Science, BonnRheinSieg University, Germany
 - Preparatory and International Phase: Graduate School Saarbrücken, CalTech, USA
 - PhD: Computer Science, Saarland University, DFKI Germany
 - Full professor for Visual Computing @ RheinMain University, Wiesbaden
- Work experience:
 - Various research projects (H-BRS, DFKI, Saarland University):
 - MoVISO (BMBF): Mobile, perception-based, real-time visualization of ISO/OGC-compliant 3D city models
 - OLIVE (BMBF): Physically-based lighting calculation to optimize stationary target spectra for intelligent, real-time control of spectrally and geometrically measured luminaires (Human-Centric Lighting)
 - PlaSMoNa: Platform for standardized integration of Web 2.0 technologies, navigation, and smartphones for social applications
 - *Dreamspace (EU): Development of real-time, high-quality rendering methods for film production*
 - ...
 - CRUSE Spezialmaschinen GmbH, Senior Software Developer
- Research interests:
 - Human Visual Perception and XR (Perception-driven graphics)
 - Eye-Tracking Techniques and Applications
 - High-quality Image Synthesis Techniques
 - GP-GPU and Parallel Computing
 - Game Technology
 - Visual Analytics



BACKGROUND

EU - GERMANY - HESSE - WIESBADEN

- Germany: one of 27 European Union States
- (Population: EU 450 mln., Germany 83 mln.)
- Hesse: one of 16 German Federal States
- Roughly at the geographical center of Germany



<https://www.sueddeutsche.de/bayern/neuer-mittelpunkt-der-eu-europas-herz-ist-eine-unterfrankische-wiese-1.1711789>



BACKGROUND

The Federal State of Hesse

I do not know what it could mean
That I am so sad;
A tale from bygone days
Will not leave my mind.

The air is cool and darkness falls,
The Rhine flows peacefully by;
The mountain's summit sparkles
In the glow of the evening sun.

The loveliest maiden sits
High above, wondrous to behold;
Her golden jewellery flashes,
And she combs her golden hair.

With a golden comb she combs it,
And sings a song as she does;
Its melody is wondrous,
Overwhelming in its power.

The boatman in his little boat
Is seized by wild despair;
He does not see the dangerous rocks—
He looks only upward to the heights.

I believe that, in the end, the waves
Swallow boatman and boat;
And that is what, through her singing,
The Lore-Ley has done.

Heinrich Heine, 1824



BACKGROUND

Quick Facts on Wiesbaden

- Capital of the federal state of Hesse
- Approx. 290,000 residents
- Founded in the Roman Age (Aquae Mattiacorum)
- The 19th century impressive architecture (Jugendstil)
- Famous theater & Kurhaus (spa) from the time of Kaiser Wilhelm II
- 14 hot springs up to 140° F (= 60° C)
- A lot of town festivals like wine festival, seasonal festivals, etc.
- European headquarters of the US Army



HSRM UNIVERSITY OF APPLIED SCIENCES

Hochschule RheinMain



- Founded in 1971 as a University of Applied Sciences
- Campus in Wiesbaden and Rüsselsheim
- ca. 12,000 students, ca. 11,000 undergraduates
- ca. 1,000 employees, > 250 professors
- Undergraduate and graduate programs:
 - Bachelor,
 - Master,
 - PhD
- **No tuition fees** for students
- Practice-driven education and applied research



Five faculties with diverse academic areas

- Design, Computer Sciences, Media (DCSM)
- Wiesbaden Business School
- Architecture & Civil Engineering
- Engineering
- Applied Social Sciences

RESEARCH

3 (4) doctoral programs, **4 research centers**, 26 research groups

Doctoral programs
with the right to
award doctorates:

Applied Computer Sciences

System-integrated Engineering
(applied for)

Mobility and Logistics

Social Work

Research centers:

SSMT

“Smart Systems for Human and
Technology”

AZARE

“Application Center for Drive Systems
Using Renewable Energy Sources (AZARE)”

RITMO

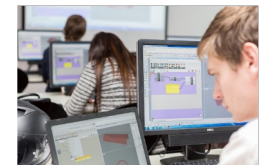
“Rhein-Main Institute for Transformative
Sciences in Mobility and Logistics”

FoRM

“RheinMain Research Center for
Professionalism in Social Work”

Research areas:

- Smart User Interface Design and Technology
- Smart Business & Digital Transformation
- Smart Tools for Software Engineering
- Learning and Visual Systems
- Internet of Smart Things
- Basic technologies
- Energy conversion technologies
- Infrastructure technologies
- Society
- Transportation
- Logistics
- Mobility
- Educational Equality and Health Literacy
- Difference & Power
- Transformative Community Development



HSRM UNIVERSITY OF APPLIED SCIENCES

Our International Network

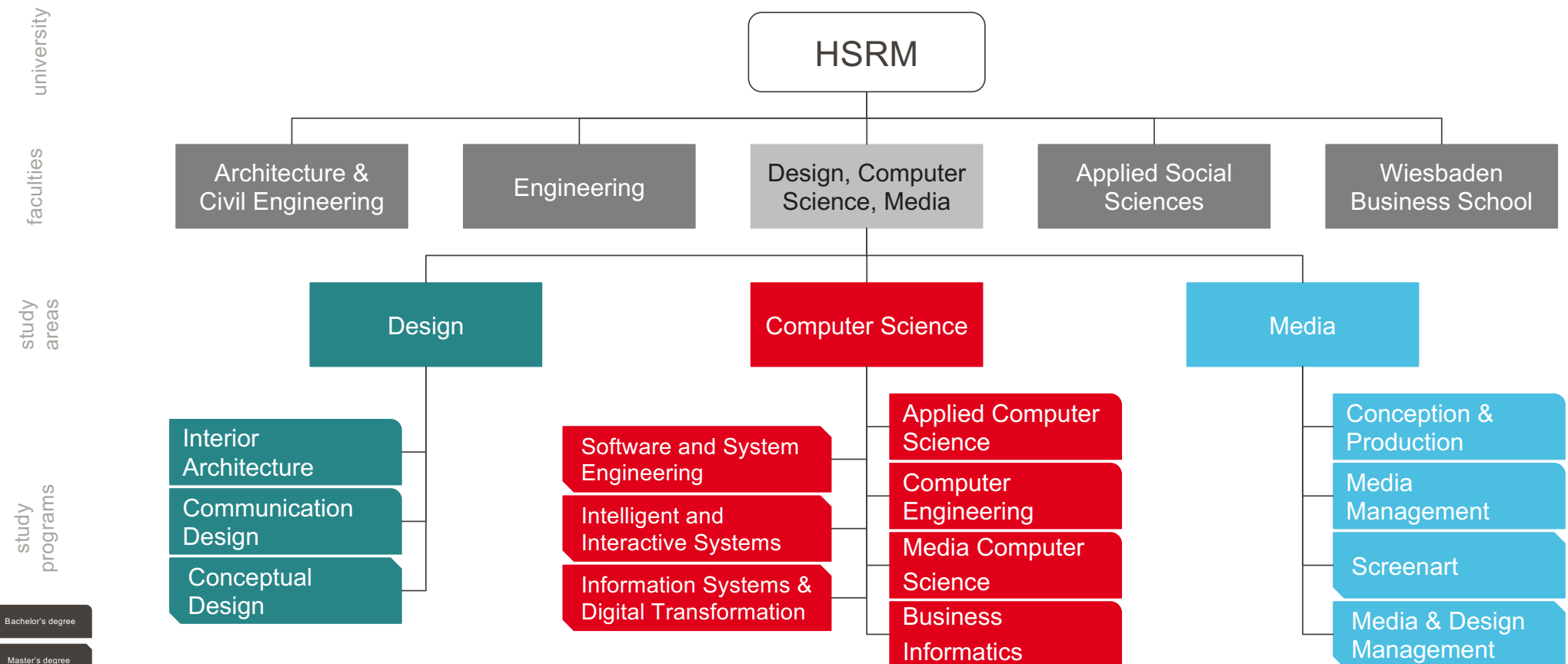


- Ca. 160 partner universities all over the world
- Students from more than 100 countries
- Active partnerships for all faculties
 - student & research programs
 - buddy programmms
 - internships
 - scholarship opportunities
- Semester & short-term programs
 - summer schools,
 - guest lecturers, ...
- We are open to new partnerships



HSRM UNIVERSITY OF APPLIED SCIENCES

Organizational structure



10.09.2025

Visit @ CareLab, NAIST - RheinMain University of Applied Sciences

CAMPUS UNTER DEN EICHEN

Campus Site in the North of Wiesbaden



RESEARCH CENTER

Smarte Systems for Mankind and Technology - SSMT

- The research center SSMT was founded 2024
- SSMT focus on timely, socially relevant intelligent IT systems
- Interdisciplinary approach members study areas of DCSM are included
- The research interests are organized in five working groups
 - **SMARTUI** Smart User Interface Design and Technology
 - **DBEG** Digital Business & E-Government
 - **IOST** Internet of Smart Things
 - **LAVIS** Learning and Visual Systems
 - **STSE** Smart Tools for Software Engineering
 - **SMARTUI** Smart User Interface Design and Technology



XR Research @ HSRM

Working Group LAVIS ("Learning and Visual Systems")



LAVIS' mission is data analysis: Our expertise covers various aspects of data analysis pipelines, from signal processing over AI to user interaction



Prof. Dr. Ralf Dörner
Computer Graphics
Visualization
[Homepage](#)



Prof. Dr.-Ing. Biying Fu
Explainable ML/AI
Data Science
[Homepage](#)



Prof. Dr. Dirk Krechel
Content Analytics
Knowledge
Management
[Homepage](#)



Prof. Dr. Ulrich Schwanecke
Computer Vision
Mixed Reality
[Homepage](#)



Prof. Dr. Adrian Ulges
Machine Learning
Natural Language
Processing
[Homepage](#)



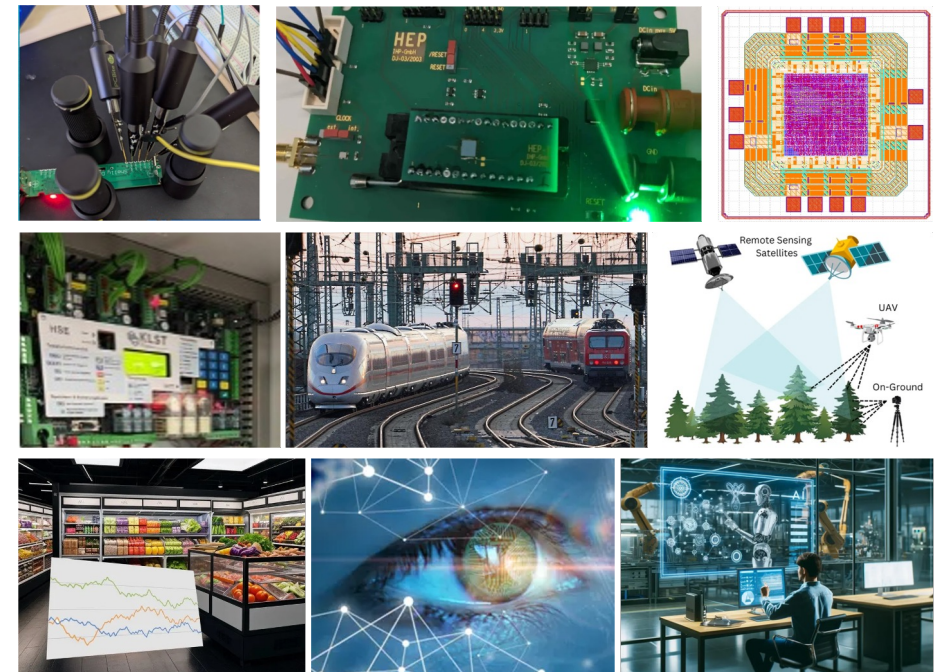
Prof. Dr.-Ing. Martin Weier
Visual Computing
Human Perception
[Homepage](#)

Research @ SSMT

Projects



- Strong research activities in AI, security & IC design
- Selection of 3rd party funded research project
 - FIENDISH - Security
 - DI-Sign-HEP - IC design/Security
 - OCDPro - IC design
 - Ankl - AI/Safety
 - QUDIS - Security
 - AI-Forcast - AI
 - IamCool - AI
 - FlowLense - AI/Security
 - VaSTNet - AI
 - StartUpLabs - startup incubator
 - TACTFUL - AI/Design



Q&A SESSION

Questions?

**Thank you very much for your
attention.**

Want to visit us, just get in touch: Martin.Weier@hs-rm.de